

Programming Assignment 2: Grover's Algorithm for Constraint Satisfaction

Spring 2026 — Assoc. Prof. Dr. Mehmet Cengiz Onbaşlı

Assignment due: 24/04/2026, 23:59

Overview

This assignment asks you to implement Grover's search algorithm *from scratch* for a concrete constraint-satisfaction problem. Unlike the weekly coding exercises, no oracle or diffusion functions are provided to you. You must design every circuit component yourself, justify each design choice in writing, and verify your implementation empirically.

You will search a space of all $N = 2^4 = 16$ four-bit strings for those satisfying the Boolean predicate

$$f(x) = (x_0 \oplus x_1) \wedge x_2,$$

where x_0 is the least significant bit (qubit 0 in Qiskit) and $\oplus$ denotes XOR. **There are exactly 4 marked strings.** Your oracle must be constructed using CNOT and Toffoli (CCX) gates together with phase kickback — not by the pattern-matching multi-controlled-Z technique from the weekly exercises.

Solutions that identify the 4 marked states and build the oracle as a list of phase-flip targets will receive **zero credit** for Part A, even if the circuit produces correct phases.

Deliverables

- **Completed Jupyter Notebook** PA2_Grover_⟨StudentID⟩.ipynb with all code cells filled in, all assertions in the auto-check cells passing, and all reflection mark-down cells completed.
Example filename: PA2_Grover_0064384.ipynb
- **Exported PDF** of the completed notebook showing all code and output. Use *File* → *Export Notebook As* → *PDF* or print-to-PDF.
- Submit both files via **KUHUB Learn** before the deadline. Do not forget to sign the Honor Code with your name and student ID inside the notebook.

Questions Summary

Part	Topic	Key Tasks	Points
A	Boolean Oracle via Phase Kickback	Build <code>build_oracle()</code> using CNOT + CCX + ancilla; verify correctness	35
B	Complete Grover Circuit	Build diffusion operator; assemble full circuit; run simulation and confirm marked states	25
C	Overshoot and Iteration Sensitivity	Compute and plot $P_{\text{theory}}(k)$ vs. $P_{\text{exp}}(k)$ for $k = 0, \dots, 5$; identify k^* and quantify overshoot	25
D	Scaling Verification	Run Grover with theoretical k^* for $n = 3, 4, 5, 6, 7$; measure empirical success probability; verify $O(\sqrt{N})$ scaling	15
Total			100

Rubric (Applied to Every Part)

Each part is graded on two equal components.

- **Working Code (50 %):** The code cell(s) for the part run without errors, all assertions in the auto-check cell pass, and the outputs are numerically correct.
- **Written Explanation (50 %):** The reflection markdown cell(s) for the part contain a clear, mathematically grounded explanation of *what* the code does, *why* it is correct, and *what* the output means. A code comment such as “apply CNOT” is not sufficient; explain the quantum-mechanical purpose of every non-trivial step.

Detailed breakdown per part:

Part	Component	Code	Explanation	Subtotal
A	Oracle construction & verification	17.5	17.5	35
B	Diffusion, full circuit, & simulation	12.5	12.5	25
C	Overshoot plots & analysis	12.5	12.5	25
D	Scaling experiment & fit	7.5	7.5	15
Total		50	50	100

Instructions

- Complete **only** the cells marked with `### YOUR CODE HERE ###` and the reflection markdown cells marked with `[YOUR EXPLANATION HERE]`.
- Do **not** modify any cell labelled **DO NOT MODIFY**.
- Before submission: restart the kernel and run all cells (*Kernel $\rightarrow$ Restart & Run All*) to ensure top-to-bottom reproducibility.
- Partial credit is awarded: a clearly written explanation of a partially working solution is valued over uncommented correct code.
- **Do not hard-code the known marked states** anywhere in your oracle. The oracle must be general, derived from the predicate logic only.

Honor Code

By submitting this assignment you affirm:

“I hereby certify that I have completed this assignment on my own without any help from anyone else. I have not used, accessed, received, or distributed any information from/to any other unauthorized source. The effort in this assignment belongs only to me.”

Sign with your full name and student ID in the first cell of the notebook.

Grading Notes

- This assignment contributes to the 20% homework component of your final grade.
- Auto-check cells use `assert` statements. A failed assertion means a numerical error in that part; re-read the specification and debug before submitting.
- Explanation quality is assessed holistically: depth of reasoning, correct use of quantum-mechanical terminology, and connection to the relevant equations and figures in the Week 7 notes.